

Methods & Overloading: AssignmentTest.java

AssignmentTest.java (Part 1):

```
// AssignmentTest.java
public class AssignmentTest {
    public static void main(String[] args) {
        // Create an instance of each class.
        Car myCar = new Car();
        Dog myDog = new Dog();
        Student myStudent = new Student();

        System.out.println("--- Testing Car Methods ---");
        // Test the Car methods.
        System.out.println("Time to drive 100 miles at 50 mph: " +
myCar.ride(100, 50) + " minutes.");
        System.out.println("Needs oil change? " + myCar.checkOil());
        System.out.println("Time to drive 5000 miles at 50 mph: " +
myCar.ride(5000, 50) + " minutes.");
        System.out.println("Needs oil change? " + myCar.checkOil());
        myCar.changeOil();
        System.out.println("Needs oil change? " + myCar.checkOil());
        System.out.println();

        System.out.println("--- Testing Dog Methods ---");
        // Test the Dog methods.
        myDog.bark();
        myDog.run();
        myDog.run();
        myDog.run();
        System.out.println();

        System.out.println("--- Testing Student Methods ---");
        // Test the Student methods.
        System.out.println("5 + 3 = " + myStudent.calculate(5, 3, '+'));
        System.out.println("10 - 2 = " + myStudent.calculate(10, 2, '-'));
        System.out.println("4 * 6 = " + myStudent.calculate(4, 6, '*'));
        System.out.println("100 / 5 = " + myStudent.calculate(100, 5, '/'));
        System.out.println();
    }
}
```

AssignmentTest (Part 2):

```
// Student.java
public class AssignmentTest {
    public static void main(String[] args) {
        System.out.println("--- Testing Overloaded Constructors and Methods  
---");

        // Overloaded Car Constructors and Methods
        Car car1 = new Car("Honda", 2022);
        Car car2 = new Car(2020);
        car1.revEngine();
        System.out.println("Time to drive 100 miles at 50 mph: " +
car1.ride(100, 50) + " minutes.");
        car2.ride(50); // Calling the overloaded ride method without speed
        System.out.println();

        // Overloaded Dog Constructors and Methods
        Dog dog1 = new Dog("Buddy", 5);
        Dog dog2 = new Dog("Max");
        dog1.bark(); // Calling default bark
        dog2.bark("WOOF!"); // Calling overloaded bark with a custom sound
        System.out.println();

        // Overloaded Student Constructors and Methods
        Student student1 = new Student("Jane", "Doe", 12345);
        Student student2 = new Student(98765);
        System.out.println("10 - 2 = " + student1.calculate(10, 2, '-')); //
Original calculate method
        System.out.println("15 + 7 = " + student2.calculate("15 + 7")); //
Overloaded calculate method with a string
        System.out.println();
    }
}
```